



2 Real Analysis 2023

Tutorial 4

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Limits of sequences

Problem 13 Let $(a_n)_{n=1}^{\infty}$ be a sequence with $\lim_{n \rightarrow \infty} a_n = 4$. Prove by definition that

$$(a) \lim_{n \rightarrow \infty} (3a_n - 2) = 10, \quad (b) \lim_{n \rightarrow \infty} a_n(a_n + 1) = 20, \quad (c) \lim_{n \rightarrow \infty} \frac{1}{a_n} = \frac{1}{4}.$$

Solution.

(a) We have

$$|(3a_n - 2) - 10| = |3a_n - 12| = 3|a_n - 4|.$$

Given $\varepsilon > 0$, it follows by the definition of the limit $\lim_{n \rightarrow \infty} a_n = 4$ that there exists $N_1 \in \mathbb{N}$ such that

$$|a_n - 4| < \frac{\varepsilon}{3} \quad \text{for all } n \geq N_1.$$

It follows that

$$|(3a_n - 2) - 10| = 3|a_n - 4| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon \quad \text{for all } n \geq N_1.$$

This proves that $\lim_{n \rightarrow \infty} (3a_n - 2) = 10$.

(b) We have

$$a_n(a_n + 1) - 20 = a_n^2 + a_n - 20 = (a_n - 4)(a_n + 5),$$

and

$$|a_n(a_n + 1) - 20| = |a_n - 4||a_n + 5|.$$

Taking $\varepsilon = 1$ in the definition of the limit $\lim_{n \rightarrow \infty} a_n = 4$, we see that there exists $N_2 \in \mathbb{N}$ such that

$$|a_n - 4| < 1 \quad \text{for all } n \geq N_2.$$

The inequality $|a_n - 4| < 1$ is equivalent to $3 < a_n < 5$. Thus, for $n \geq N_2$ we have

$$|a_n(a_n + 1) - 20| = |a_n - 4||a_n + 5| < 10|a_n - 4|.$$

Given $\varepsilon > 0$, there is $N_3 \in \mathbb{N}$ such that

$$|a_n - 4| < \frac{\varepsilon}{10} \quad \text{for all } n \geq N_1.$$

Then for all $n \geq \max\{N_2, N_3\}$ we have

$$|a_n(a_n + 1) - 20| = |a_n - 4||a_n + 5| < 10|a_n - 4| < 10 \cdot \frac{\varepsilon}{10} = \varepsilon.$$

This shows that $\lim_{n \rightarrow \infty} a_n(a_n + 1) = 20$.

(c) We have

$$\left| \frac{1}{a_n} - \frac{1}{4} \right| = \frac{|a_n - 4|}{4|a_n|}.$$

With $N_2 \in \mathbb{N}$ like in (b), we have that $3 < a_n < 5$ for all $n \geq N_2$. The inequality $3 < a_n < 5$ is equivalent to $\frac{1}{5} < \frac{1}{a_n} < \frac{1}{3}$. Thus, for $n \geq N_2$ we have

$$\left| \frac{1}{a_n} - \frac{1}{4} \right| = \frac{|a_n - 4|}{4|a_n|} < \frac{|a_n - 4|}{12} < |a_n - 4|.$$

Given $\varepsilon > 0$, there is $N_0 \in \mathbb{N}$ such that

$$|a_n - 4| < \varepsilon \quad \text{for all } n \geq N_0.$$

Then for all $n \geq \max\{N_2, N_0\}$ we have

$$\left| \frac{1}{a_n} - \frac{1}{4} \right| < |a_n - 4| < \varepsilon.$$

This proves that $\lim_{n \rightarrow \infty} \frac{1}{a_n} = \frac{1}{4}$.

Problem 14 Prove by definition that $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$.

Solution.

We have

$$\begin{aligned} |(\sqrt{n+1} - \sqrt{n}) - 0| &= \sqrt{n+1} - \sqrt{n} = \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}}. \end{aligned}$$

Let $\varepsilon > 0$. We want to have $\frac{1}{2\sqrt{n}} < \varepsilon$. Modifying this inequality, we come to equivalent inequalities $\sqrt{n} > \frac{1}{2\varepsilon}$, $n > \frac{1}{4\varepsilon^2}$.

We finish the proof as follows. Given $\varepsilon > 0$, take $N \in \mathbb{N}$ with $N > \frac{1}{4\varepsilon^2}$ (such N exists by the Archimedean Property). Then for any $n \geq N$ we have

$$|(\sqrt{n+1} - \sqrt{n}) - 0| < \frac{1}{2\sqrt{n}} < \frac{1}{2\sqrt{N}} < \varepsilon.$$

This proves that $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$.

Problem 15 Use the Squeeze Theorem and rules for working with limits to calculate the following limits:

$$(a) \lim_{n \rightarrow \infty} \frac{n \cos(n^2)}{2n^2 + 1}, \quad (b) \lim_{n \rightarrow \infty} \frac{1 + (-1)^n}{n}.$$

Solution.

(a) We have $-1 \leq \cos(n^2) \leq 1$, and thus

$$\frac{-n}{2n^2 + 1} \leq \frac{n \cos(n^2)}{2n^2 + 1} \leq \frac{n}{2n^2 + 1}.$$

Since

$$\lim_{n \rightarrow \infty} \frac{n}{2n^2 + 1} = \lim_{n \rightarrow \infty} \frac{1}{n + \frac{1}{n}} = 0$$

and

$$\lim_{n \rightarrow \infty} \frac{-n}{2n^2 + 1} = \lim_{n \rightarrow \infty} \frac{-1}{n + \frac{1}{n}} = 0,$$

we conclude by the Squeeze Theorem that

$$\lim_{n \rightarrow \infty} \frac{n \cos(n^2)}{2n^2 + 1} = 0.$$

(b) We have $-1 \leq (-1)^n \leq 1$, and thus

$$0 \leq \frac{1 + (-1)^n}{n} \leq \frac{2}{n}.$$

Since $\lim_{n \rightarrow \infty} \frac{2}{n} = 0$ (and, clearly, $\lim_{n \rightarrow \infty} 0 = 0$), the Squeeze Theorem implies

$$\lim_{n \rightarrow \infty} \frac{1 + (-1)^n}{n} = 0.$$

Problem 16 Recall that a sequence $(a_n)_{n=1}^{\infty}$ is called *eventually constant* if there are $a \in \mathbb{R}$ and $N \in \mathbb{N}$ such that $a_n = a$ for all $n \geq N$.

Prove or disprove:

- (a) Every eventually constant sequence is convergent.
- (b) Every convergent sequence is eventually constant.
- (c) Every eventually constant sequence is bounded.

Solution.

(a) True. Assume that $a_n = a$ for all $n \geq N$. Then for any $\varepsilon > 0$ we have

$$|a_n - a| = 0 < \varepsilon \quad \text{for all } n \geq N.$$

This proves that $\lim_{n \rightarrow \infty} a_n = a$.

(b) False. For example, the sequence $\left(\frac{1}{n}\right)_{n=1}^{\infty}$ is convergent with limit 0, but it is not eventually constant.

(c) This is true by the facts that every eventually constant sequence is convergent, and every convergent sequence is bounded. This can be also easily seen directly as

$$|a_n| \leq \max\{|a_1|, |a_2|, \dots, |a_N|\} \quad \text{for all } n \in \mathbb{N}.$$